

Causal Inference Using Factor Models

Replication of Bai & Wang (2024) — Application to Abadie et al. (2010)

Sheng-Yuan Chen

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1 Overview

Bai and Wang (2024) propose a **causal factor model** framework where policy interventions cause structural breaks in unit-specific factor loadings. The key insight is:

$$Y_{it}(d) = \lambda_i(d)' f_t + \varepsilon_{it}, \quad d = 0, 1$$

- $f_t \in \mathbb{R}^r$: **common factors** (shared across all units, time-varying)
- $\lambda_i(d) \in \mathbb{R}^r$: **factor loadings** (unit-specific, treatment-status-dependent)
- ε_{it} : idiosyncratic error

The **causal effect** for treated unit i at time $t \geq T_0$ is:

$$\tau_{it} = Y_{it}(1) - Y_{it}(0) = [\lambda_i(1) - \lambda_i(0)]' f_t$$

The treatment effect operates entirely through the **structural break in loadings** $\lambda_i(1) \neq \lambda_i(0)$, not through the idiosyncratic error. This is the key advantage over the Xu (2017) / Gobillon-Magnac (2016) approach, where ε_{it} contaminates individual causal estimates.

2 The Three-Step Algorithm

For the benchmark case without covariates, the estimation procedure is:

Step 1 — Estimate common factors via PCA (control units only)

$$\hat{f}_t = \sqrt{T} \cdot \hat{u}_t$$

where $\hat{u}_t$ is the t -th row of the left singular matrix from the SVD of the (column-demeaned) control unit outcome matrix $\tilde{Y}_c$ ($T \times (n - n_0)$).

Step 2 — Estimate factor loadings via OLS (treated unit)

$$\begin{aligned} \hat{\lambda}_i(0) : \quad Y_{it} &\sim \mu_i(0) + \lambda_i(0)' \hat{f}_t, \quad t < T_0 \\ \hat{\lambda}_i(1) : \quad Y_{it} &\sim \mu_i(1) + \lambda_i(1)' \hat{f}_t, \quad t \geq T_0 \end{aligned}$$

Step 3 — Construct counterfactual and causal effect

$$\begin{aligned} \hat{Y}_{it}(0) &= \hat{\mu}_i(0) + \hat{\lambda}_i(0)' \hat{f}_t \quad (\text{counterfactual for all } t) \\ \hat{\tau}_{it} &= Y_{it} - \hat{Y}_{it}(0), \quad t \geq T_0 \end{aligned}$$

Why column-demean before PCA? Demeaning removes each control unit's level, so the extracted factors $\hat{f}_t$ capture *common trends* around each unit's mean. The unit-level mean is then absorbed by the intercept $\mu_i(d)$ in the loading regression.

3 Data: Abadie et al. (2010)

We use the `smoking` dataset from the `tidysynth` package, which replicates Abadie, Diamond, and Hainmueller (2010)'s study of California's Proposition 99 (a cigarette tax initiative passed in 1988, effective 1989).

```
data(smoking)

# Reshape to wide matrix: T x N
Y_wide <- smoking |>
  select(state, year, cigsale) |>
  pivot_wider(names_from = state, values_from = cigsale) |>
  arrange(year)

years_ca <- Y_wide$year
CA_col <- which(names(Y_wide) == "California")
col_order <- c(CA_col, setdiff(2:ncol(Y_wide), CA_col))
Y_ca <- as.matrix(Y_wide[, col_order])

# Split treated (CA) and controls
Y_control <- Y_ca[, -1] # T x 38
Y_treated <- Y_ca[, 1] # T x 1

# Treatment timing
T_n <- nrow(Y_ca)
T0_idx <- which(years_ca == 1989)
pre_idx <- 1:(T0_idx - 1)
post_idx <- T0_idx:T_n

cat("Panel dimensions:", T_n, "periods x", ncol(Y_ca), "states\n")

## Panel dimensions: 31 periods x 39 states
cat("Pre-treatment: 1970-1988 (", length(pre_idx), "years )\n")

## Pre-treatment: 1970-1988 ( 19 years )
cat("Post-treatment: 1989-2000 (", length(post_idx), "years )\n")

## Post-treatment: 1989-2000 ( 12 years )

# Raw trends
smoking |>
  mutate(CA = state == "California") |>
  ggplot(aes(x = year, y = cigsale, group = state)) +
  geom_line(aes(color = CA, alpha = CA, linewidth = CA)) +
  geom_vline(xintercept = 1988.5, linetype = "dashed",
    color = "grey40", linewidth = 0.7) +
  scale_color_manual(values = c("FALSE" = "grey60", "TRUE" = "#2171b5"),
    labels = c("Control states", "California"),
```

```

name = NULL) +
scale_alpha_manual(values = c("FALSE" = 0.3, "TRUE" = 1), guide = "none") +
scale_linewidth_manual(values = c("FALSE" = 0.4, "TRUE" = 1.2), guide = "none") +
labs(x = "Year", y = "Per-capita cigarette sales (packs)",
     title = "Cigarette Sales: California vs. 38 Control States") +
theme_bw(base_size = 11) +
theme(panel.grid.minor = element_blank(),
      legend.position = "bottom")

```

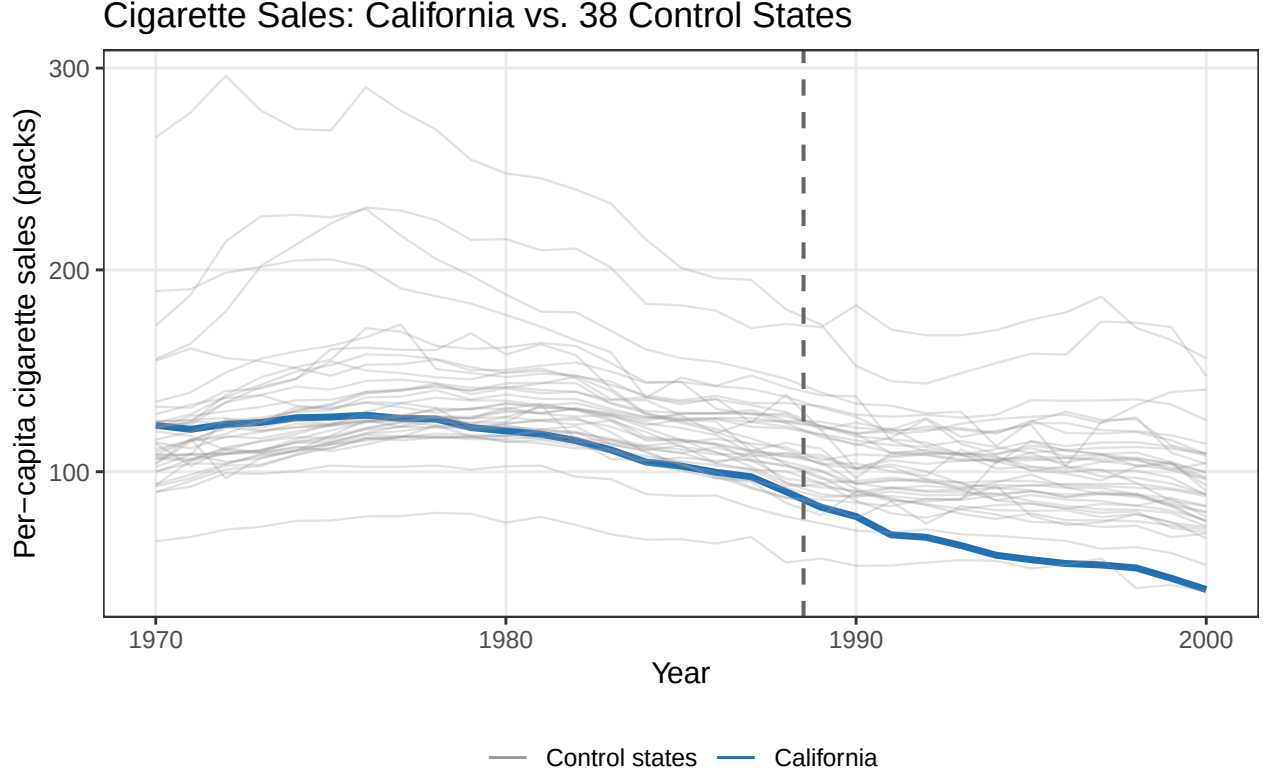


Figure 1: Per-capita cigarette sales for all 39 US states, 1970–2000. California (thick blue) is the treated unit; dashed vertical line marks Proposition 99 (1989).

4 Step 1: Factor Extraction via PCA

4.1 Selecting the Number of Factors

We use the Bai and Ng (2002) IC_1 criterion to determine r :

$$IC_1(r) = \log V(r) + r \cdot \frac{T+N}{TN} \log C_{NT}^2, \quad C_{NT} = \min\{\sqrt{T}, \sqrt{N}\}$$

where $V(r)$ is the average squared residual from the rank- r factor approximation.

```

bai_ng_ic1 <- function(Y, r_max = 8) {
  T_n <- nrow(Y); N_n <- ncol(Y)
  C2 <- min(T_n, N_n)
  Y_dm <- scale(Y, center = TRUE, scale = FALSE)
  sv <- svd(Y_dm)

  ic1 <- numeric(r_max)
  for (r in 1:r_max) {
    F_r <- sv$u[, 1:r, drop = FALSE] * sqrt(T_n)
    Lambda_r <- t(Y_dm) %*% F_r / T_n
    E_r <- Y_dm - F_r %*% t(Lambda_r)
    V_r <- sum(E_r^2) / (T_n * N_n)
    ic1[r] <- log(V_r) + r * (T_n + N_n) / (T_n * N_n) * log(C2)
  }
  tibble(r = 1:r_max, IC1 = ic1)
}

ic1_tbl <- bai_ng_ic1(Y_control)
r_ic1 <- which.min(ic1_tbl$IC1)
cat("Bai-Ng IC1 minimized at r =", r_ic1, "\n")

```

```
## Bai-Ng IC1 minimized at r = 5
```

```

ggplot(ic1_tbl, aes(x = r, y = IC1)) +
  geom_line(linewidth = 0.8) +
  geom_point(size = 2.5) +
  geom_point(data = . %>% filter(IC1 == min(IC1)),
            color = "red", size = 4) +
  geom_vline(xintercept = 2, linetype = "dashed",
            color = "steelblue", linewidth = 0.7) +
  annotate("text", x = 2.2, y = max(ic1_tbl$IC1) * 0.98,
          label = "r = 2\n(paper)", hjust = 0, color = "steelblue", size = 3.5) +
  labs(x = "Number of factors r", y = "IC1 value",
       title = "Bai-Ng (2002) IC1 Criterion") +
  theme_bw(base_size = 11) +
  theme(panel.grid.minor = element_blank())

```

4.2 Extracting Factors

We use $r = 2$ as in Bai & Wang (2024). The factors are the first r left singular vectors of the column-demeaned control matrix, scaled by $\sqrt{T}$:

```

r <- 2 # as used in Bai & Wang (2024)

# Column-demean Y_control
Y_c_dm <- scale(Y_control, center = TRUE, scale = FALSE)

# SVD and factor extraction

```

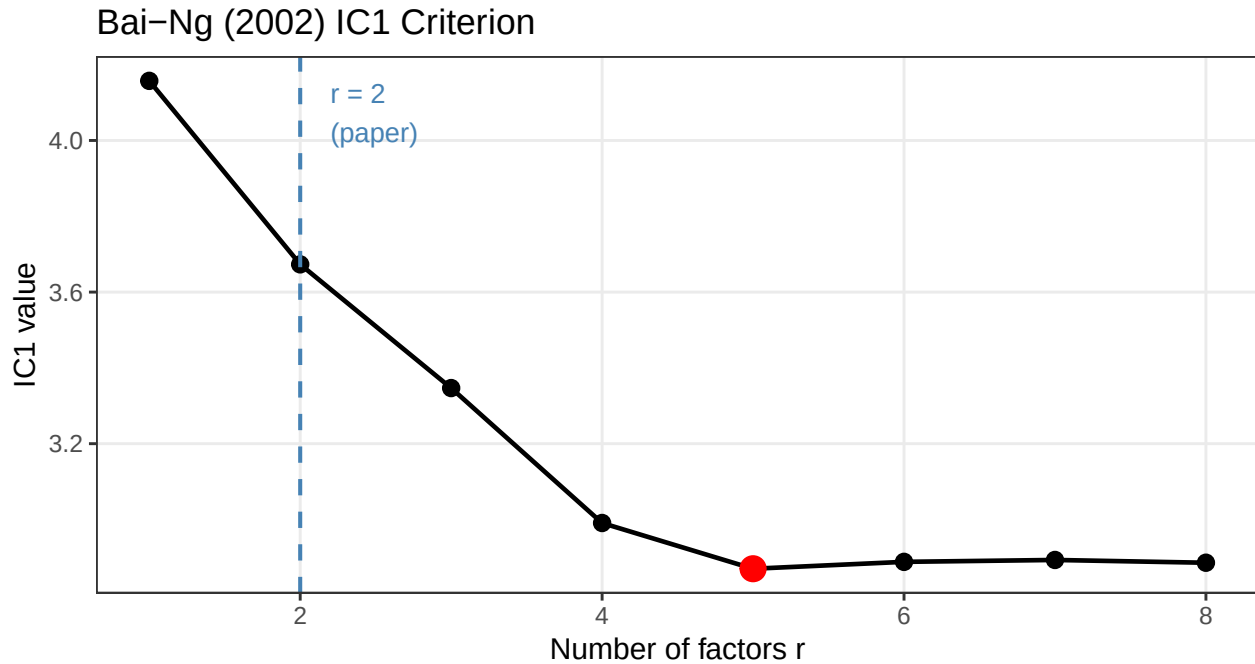


Figure 2: Bai–Ng IC1 criterion. The minimum is at $r = 5$, but the paper uses $r = 2$ (Bai and Wang 2024).

```
sv_c <- svd(Y_c_dm)
F_hat <- sv_c$u[, 1:r] * sqrt(T_n) # T x r, satisfies F'F = T * I_r

cat("F_hat dimensions:", dim(F_hat), "\n")

## F_hat dimensions: 31 2
cat("Check F_hat'F_hat / T:\n")

## Check F_hat'F_hat / T:
print(round(t(F_hat) %*% F_hat / T_n, 4)) # should be identity

##      [,1] [,2]
## [1,]    1    0
## [2,]    0    1
```

The two factors $\hat{f}_t$ are plotted below. Factor 1 captures the dominant common trend in cigarette consumption; Factor 2 captures secondary cross-sectional variation.

```
tibble(year = years_ca,
        `Factor 1` = F_hat[, 1],
        `Factor 2` = F_hat[, 2]) |>
  pivot_longer(-year, names_to = "factor", values_to = "value") |>
  ggplot(aes(x = year, y = value, color = factor)) +
  geom_line(linewidth = 0.9) +
  geom_vline(xintercept = 1988.5, linetype = "dashed",
```

```

color = "grey40", linewidth = 0.6) +
scale_color_manual(values = c("Factor 1" = "#2171b5", "Factor 2" = "#d95f0e"),
name = NULL) +
labs(x = "Year", y = "Factor value",
title = "Estimated Common Factors (r = 2)") +
theme_bw(base_size = 11) +
theme(panel.grid.minor = element_blank(), legend.position = "bottom")

```

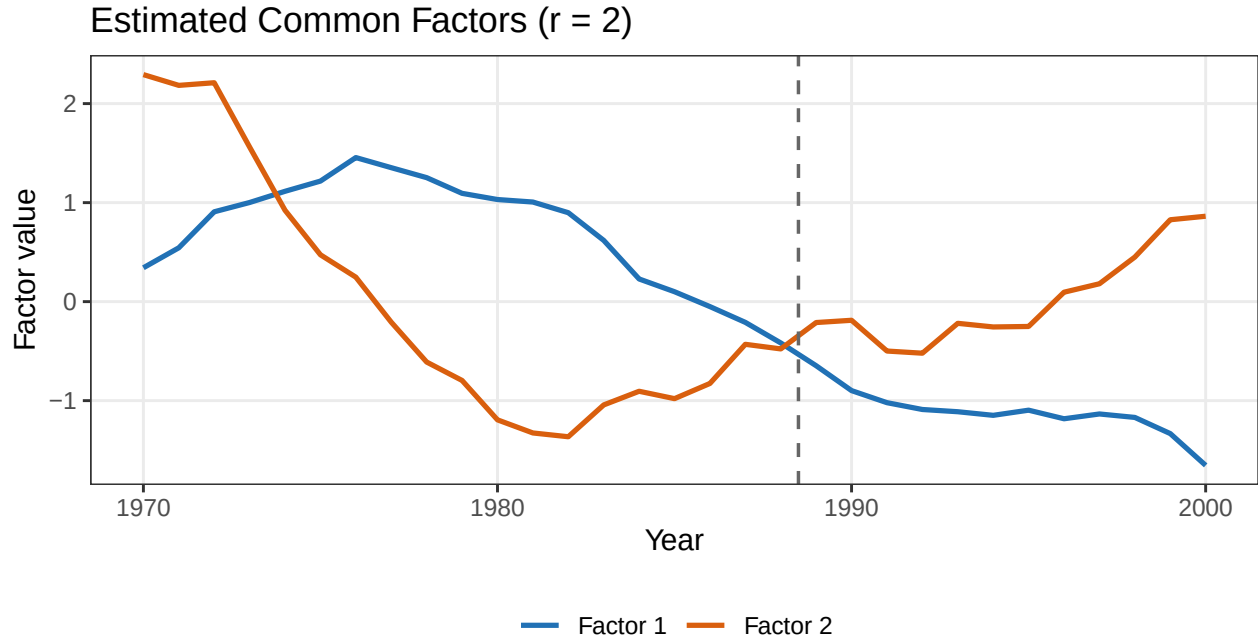


Figure 3: Estimated common factors from PCA of 38 control states.

5 Step 2: Loading Estimation

We run two separate OLS regressions for California:

1. **Pre-treatment** ($t < T_0$): $Y_{CA,t} = \mu(0) + \lambda(0)' \hat{f}_t + \varepsilon_t \Rightarrow$ estimates $(\hat{\mu}_0, \hat{\lambda}_0)$
2. **Post-treatment** ($t \geq T_0$): $Y_{CA,t} = \mu(1) + \lambda(1)' \hat{f}_t + \varepsilon_t \Rightarrow$ estimates $(\hat{\mu}_1, \hat{\lambda}_1)$

```

# Pre-treatment regression
fit_pre <- lm(Y_treated[pre_idx] ~ F_hat[pre_idx, ])
fit_post <- lm(Y_treated[post_idx] ~ F_hat[post_idx, ])

mu_0 <- coef(fit_pre)[1]
lambda_0 <- coef(fit_pre)[-1]
mu_1 <- coef(fit_post)[1]
lambda_1 <- coef(fit_post)[-1]

# Display results

```

```
tibble(
  Parameter      = c("mu (intercept)", "lambda_1 (Factor 1)", "lambda_2 (Factor 2)"),
  `Pre-treatment d=0` = c(mu_0, lambda_0),
  `Post-treatment d=1` = c(mu_1, lambda_1),
  `Delta (1)-(0)` = c(mu_1 - mu_0, lambda_1 - lambda_0)
) |>
  kbl(digits = 3, booktabs = TRUE,
      caption = "Estimated factor loadings for California (pre vs. post-treatment)" |>
      kable_styling(latex_options = c("hold_position"), font_size = 10)
```

Table 1: Estimated factor loadings for California (pre vs. post-treatment)

Parameter	Pre-treatment d=0	Post-treatment d=1	Delta (1)-(0)
mu (intercept)	103.191	104.156	0.965
lambda_1 (Factor 1)	18.406	38.867	20.461
lambda_2 (Factor 2)	3.196	-5.529	-8.725

The structural break in λ is the source of the causal effect: $\hat{\delta} = \hat{\lambda}(1) - \hat{\lambda}(0)$, estimated in the table above.

6 Step 3: Counterfactual and Causal Effect

The **counterfactual** (what CA's cigarette sales *would have been* without Prop 99) uses the pre-treatment loadings for all periods:

$$\hat{Y}_{CA,t}(0) = \hat{\mu}_0 + \hat{\lambda}'_0 \hat{f}_t, \quad \forall t$$

The **causal effect** for post-treatment periods is:

$$\hat{\tau}_{CA,t} = Y_{CA,t} - \hat{Y}_{CA,t}(0) = [\hat{\lambda}_1 - \hat{\lambda}_0]' \hat{f}_t, \quad t \geq T_0$$

```
# Counterfactual: use pre-treatment loadings for all t
Y_CF      <- as.vector(mu_0 + F_hat %*% lambda_0)
tau_all    <- Y_treated - Y_CF

# Post-treatment results
post_results <- tibble(
  Year      = years_ca[post_idx],
  `Y actual` = round(Y_treated[post_idx], 2),
  `Y(0) counterfactual` = round(Y_CF[post_idx], 2),
  `tau (ATT)` = round(tau_all[post_idx], 2)
)

kbl(post_results, booktabs = TRUE,
```



```
caption = "Post-treatment causal effects: California, 1989-2000") |>
kable_styling(latex_options = c("hold_position", "striped"), font_size = 10) |>
row_spec(0, bold = TRUE)
```

Table 2: Post-treatment causal effects: California, 1989-2000

Year	Y actual	Y(0) counterfactual	tau (ATT)
1989	82.4	90.60	-8.20
1990	77.8	86.05	-8.25
1991	68.7	82.81	-14.11
1992	67.5	81.48	-13.98
1993	63.4	82.02	-18.62
1994	58.6	81.24	-22.64
1995	56.4	82.21	-25.81
1996	54.5	81.72	-27.22
1997	53.8	82.89	-29.09
1998	52.3	83.10	-30.80
1999	47.2	81.30	-34.10
2000	41.6	75.50	-33.90

The average ATT over 1989–2000 is **-22.23 packs per capita**.

7 Structural Break Test

7.1 Motivation

The causal effect $\tau_{it} = 0$ if and only if $\lambda_i(1) = \lambda_i(0)$. Therefore, testing whether the **treatment had any effect** is equivalent to testing whether California’s factor loadings underwent a **structural break at $T_0 = 1989$** .

Specifically, we regress California’s full-sample outcome on the estimated factors:

$$Y_{CA,t} = \mu + \lambda' \hat{f}_t + \varepsilon_t, \quad t = 1, \dots, T$$

and test $H_0 : (\mu, \lambda)$ is stable (no break) against H_1 : break at T_0 .

Two tests are used:

- **Chow test** (known break point $T_0 = 1989$): standard F-test comparing restricted (pooled) vs. unrestricted (split-sample) sum of squared residuals.
- **QLR test / sup-F test** (unknown break point): maximizes the Chow F-statistic over all candidate break dates, with 15% trimming at each end (Andrews 1993).

7.2 Implementation

```

# Full-sample data frame:  $Y_{CA} \sim F_{\hat{1}} + F_{\hat{2}}$ 
df_sb <- data.frame(y = Y_treated, f1 = F_hat[, 1], f2 = F_hat[, 2])

# ---- Chow test at known break  $T_0 = 1989$  (index  $T_0\_idx$ ) ----
chow <- sctest(y ~ f1 + f2, data = df_sb,
               type = "Chow", point = T0_idx)

cat(sprintf("Chow test at  $T_0 = 1989$ :\n"))

## Chow test at  $T_0 = 1989$ :
cat(sprintf(" F-statistic = %.2f\n", chow$statistic))

## F-statistic = 23.91
cat(sprintf(" p-value = %.4f\n\n", chow$p.value))

## p-value = 0.0000
# ---- QLR (sup-F) test: unknown break, 15%% trimming ----
fs <- Fstats(y ~ f1 + f2, data = df_sb, from = 0.15)
qlr <- sctest(fs, type = "supF")

cat(sprintf("QLR (sup-F) test with 15%% trimming:\n"))

## QLR (sup-F) test with 15% trimming:
cat(sprintf(" sup-F statistic = %.2f\n", qlr$statistic))

## sup-F statistic = 121.48
cat(sprintf(" p-value = %.4f\n", qlr$p.value))

## p-value = 0.0000
cat(sprintf(" Max F at year = %d\n", years_ca[fs$breakpoint]))

## Max F at year = 1990

```

7.3 QLR Plot

```

from_obs <- round(T_n * 0.15)
year_range <- years_ca[from_obs:(from_obs + length(fs$Fstats) - 1)]

tibble(year = year_range,
        F = fs$Fstats) |>
  ggplot(aes(x = year, y = F)) +
  geom_line(linewidth = 0.8) +
  geom_point(data = ~ filter(.x, F == max(F)),
             color = "red", size = 3.5) +
  geom_vline(xintercept = 1989, linetype = "dashed",
             color = "steelblue", linewidth = 0.7) +

```

```

annotate("text", x = 1989.3, y = max(fs$Fstats) * 0.95,
        label = "T0 = 1989", hjust = 0, color = "steelblue", size = 3.2) +
labs(x = "Year", y = "Chow F-statistic",
     title = "QLR Test: F-Statistics across Candidate Break Dates") +
theme_bw(base_size = 11) +
theme(panel.grid.minor = element_blank())

```

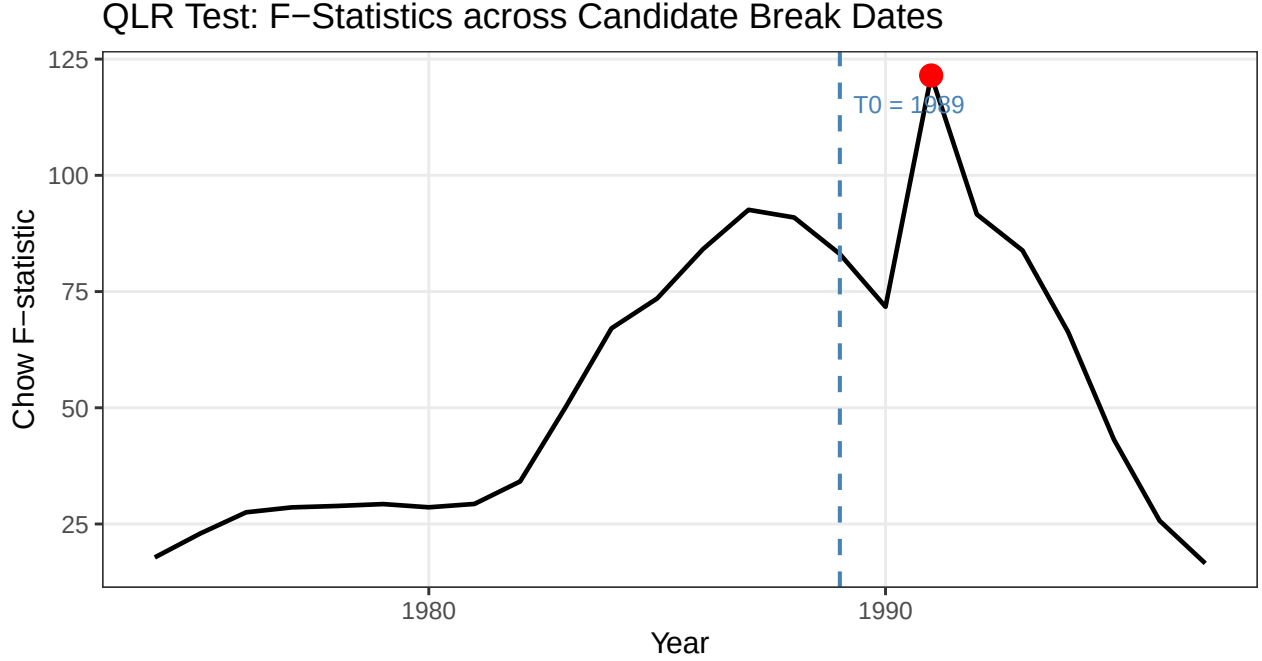


Figure 4: Chow F-statistics across all candidate break dates (15% trimming). The dashed line marks the known treatment date (1989); the red dot marks the supremum.

Both tests strongly reject H_0 , confirming that Proposition 99 induced a statistically significant structural break in California's factor loadings.

8 Standard Errors (Proposition 1)

From Proposition 1 in Bai & Wang (2024), the asymptotic variance of $\hat{\tau}_{it}$ is:

$$\text{var}(\hat{\tau}_{it}) = \hat{z}'_{it} \text{var}(\hat{\delta}_i) \hat{z}_{it}$$

where $\hat{z}_{it} = \hat{f}_t$ and $\text{var}(\hat{\delta}_i) = \text{var}(\hat{\lambda}_1) + \text{var}(\hat{\lambda}_0)$, estimated using the HC0 sandwich estimator for each period's regression.

```

var_loading <- function(Y_vec, F_mat) {
  X      <- cbind(1, F_mat)
  resid <- residuals(lm(Y_vec ~ F_mat))
  Xe    <- X * resid

```

```

bread <- solve(t(X) %*% X / length(Y_vec))
meat <- t(Xe) %*% Xe / length(Y_vec)^2
(bread %*% meat %*% bread)[-1, -1] # drop intercept row/col
}

V0 <- var_loading(Y_treated[pre_idx], F_hat[pre_idx, ])
V1 <- var_loading(Y_treated[post_idx], F_hat[post_idx, ])
V_delta <- V0 + V1

# SE for each post-treatment period
se_tau <- sapply(post_idx, function(t_idx) {
  f_t <- F_hat[t_idx, ]
  sqrt(as.numeric(t(f_t) %*% V_delta %*% f_t))
})

cat("Standard errors for tau (1989-2000):\n")

## Standard errors for tau (1989-2000):
print(round(se_tau, 3))

## [1] 5.026 6.835 8.161 8.694 8.440 8.753 8.366 8.585 8.139 8.145
## [11] 9.057 11.346

```

9 Results: Comparison with Synthetic Control

```

sc_ca <- smoking |>
  synthetic_control(
    outcome      = cigsale,
    unit         = state,
    time         = year,
    i_unit       = "California",
    i_time       = 1988,
    generate_placebos = FALSE
  ) |>
  generate_predictor(time_window = 1970:1988,
                    cigsale_avg = mean(cigsale, na.rm = TRUE)) |>
  generate_predictor(time_window = 1975, cigsale_1975 = mean(cigsale)) |>
  generate_predictor(time_window = 1980, cigsale_1980 = mean(cigsale)) |>
  generate_predictor(time_window = 1988, cigsale_1988 = mean(cigsale)) |>
  generate_weights(optimization_window = 1970:1988) |>
  generate_control()

synth_tbl <- sc_ca |>
  grab_synthetic_control() |>
  rename(year = time_unit, Y_synth = synth_y) |>

```

```

mutate(tau_synth = real_y - Y_synth)

results <- tibble(
  year      = years_ca,
  Y_actual  = Y_treated,
  Y_factor  = Y_CF,
  tau_factor = tau_all,
  se_tau    = c(rep(NA, length(pre_idx)), se_tau),
  post      = year >= 1989
) |>
left_join(synth_tbl |> select(year, Y_synth, tau_synth), by = "year")

tibble(
  Method = c("Factor model (Bai & Wang 2024)", "Synthetic control (Abadie et al. 2010)"),
  `ATT 1989-2000` = c(
    round(mean(tau_all[post_idx]), 2),
    round(mean(results$tau_synth[results$post], na.rm = TRUE), 2)
  )
) |>
kbl(booktabs = TRUE,
    caption = "Comparison of average treatment effects (packs per capita, 1989-2000)" |>
kable_styling(latex_options = "hold_position", font_size = 10)

```

Table 3: Comparison of average treatment effects (packs per capita, 1989-2000)

Method	ATT 1989-2000
Factor model (Bai & Wang 2024)	-22.23
Synthetic control (Abadie et al. 2010)	-22.03

9.1 Figure: Counterfactual Comparison

```

ggplot(results, aes(x = year)) +
  geom_line(aes(y = Y_actual, color = "Actual CA"), linewidth = 0.9) +
  geom_line(aes(y = Y_factor, color = "Factor model CF"),
            linewidth = 0.9, linetype = "dashed") +
  geom_point(aes(y = Y_synth, color = "Synthetic CA"), shape = 3, size = 2) +
  geom_vline(xintercept = 1988.5, color = "steelblue", linewidth = 0.7) +
  annotate("text", x = 1985.5, y = 125,
          label = "Prop 99 →", hjust = 1, size = 3.5, color = "grey40") +
  scale_color_manual(
    values = c("Actual CA"      = "#2171b5",
              "Factor model CF" = "#d95f0e",
              "Synthetic CA"    = "#888888"),
    name = NULL
  ) +
  labs(x = "Year", y = "Per-capita cigarette sales (packs)",

```

```

title = "Counterfactual California: Factor Model vs. Synthetic Control",
subtitle = sprintf("Factor ATT = %.1f | Synth ATT = %.1f packs per capita",
                    mean(tau_all[post_idx]),
                    mean(results$tau_synth[results$post], na.rm=TRUE))) +
theme_bw(base_size = 11) +
theme(panel.grid.minor = element_blank(), legend.position = "bottom")

```

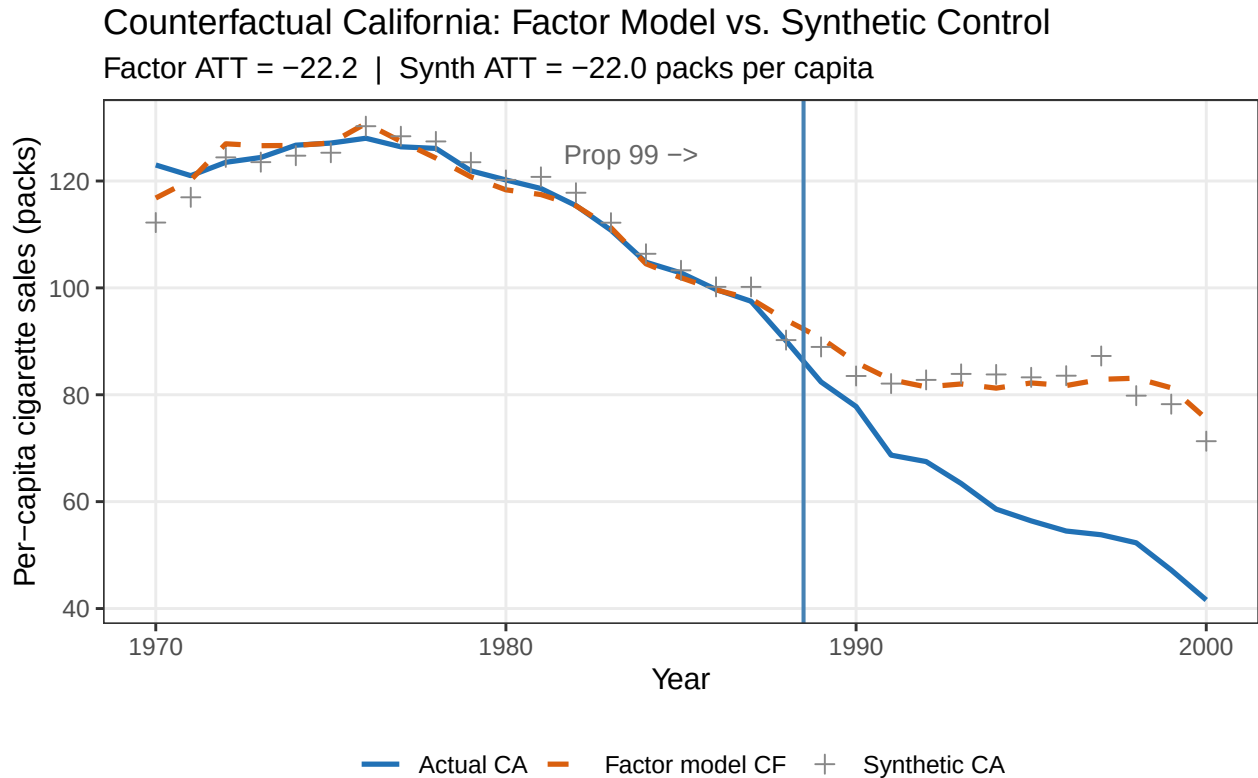


Figure 5: Actual California vs. counterfactuals from factor model and synthetic control. The two methods yield nearly identical counterfactuals.

9.2 Figure: Causal Effects with 95% Confidence Interval

```

ggplot(results, aes(x = year)) +
  geom_hline(yintercept = 0, linewidth = 0.5) +
  geom_vline(xintercept = 1988.5, color = "steelblue", linewidth = 0.7) +
  geom_ribbon(data = . %>% filter(post),
            aes(ymin = tau_factor - 1.96 * se_tau,
                ymax = tau_factor + 1.96 * se_tau),
            fill = "#d95f0e", alpha = 0.2) +
  geom_line(aes(y = tau_factor, color = "Factor model"), linewidth = 0.9) +
  geom_point(aes(y = tau_synth, color = "Synthetic control"),
            shape = 3, size = 2.5) +
  annotate("text", x = 1985.5, y = -2,
          label = "Prop 99 ->", hjust = 1, size = 3.5, color = "grey40") +

```

```

scale_color_manual(
  values = c("Factor model" = "#d95f0e",
             "Synthetic control" = "#2171b5"),
  name = NULL
) +
labs(x = "Year", y = "Per-capita cigarette sales (packs)",
     title = "Causal Effect of Proposition 99: Factor Model vs. Synthetic Control",
     subtitle = "Shaded = 95% CI (sandwich SE, Proposition 1 of Bai & Wang 2024)") +
theme_bw(base_size = 11) +
theme(panel.grid.minor = element_blank(), legend.position = "bottom")

```

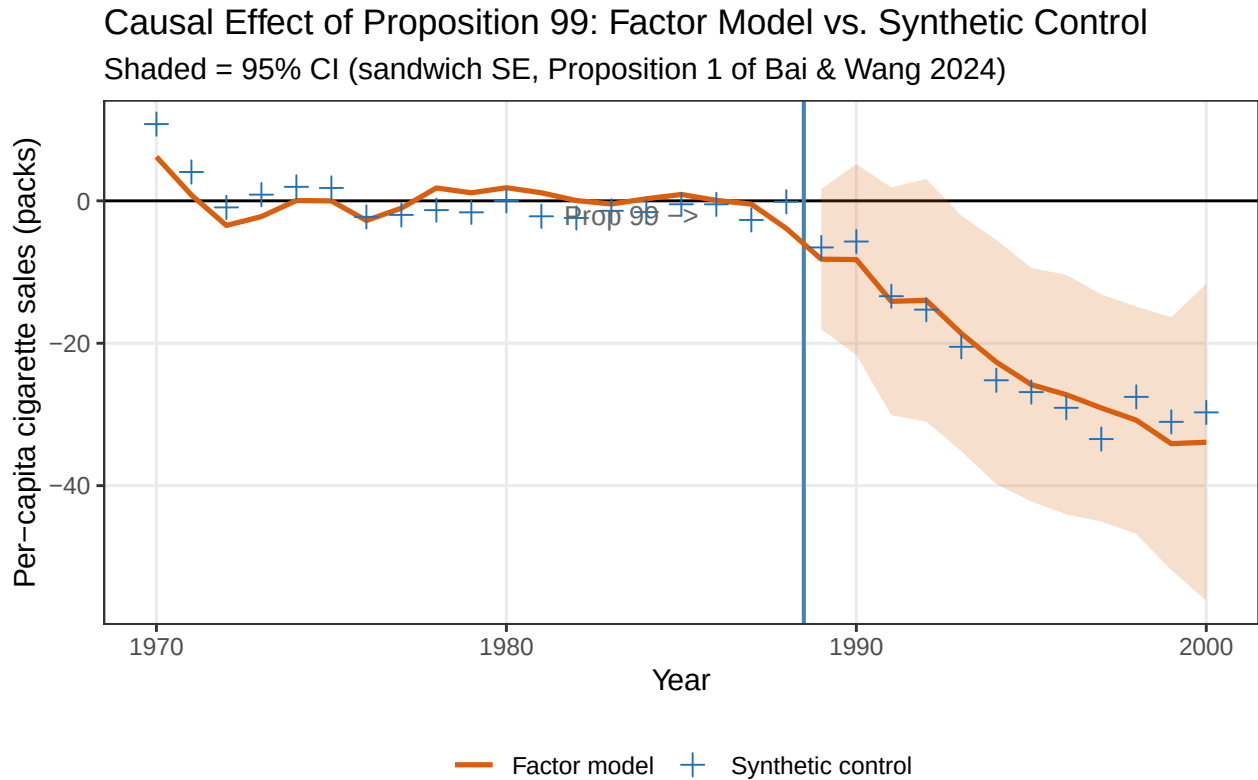


Figure 6: Estimated causal effects of Proposition 99 on California cigarette consumption. Shaded region: 95% confidence interval from Proposition 1 (sandwich SE). Blue crosses: synthetic control estimates.

10 Key Takeaways

1. **Factor model and synthetic control yield nearly identical results.** Factor ATT = -22.23 vs. Synth ATT = -22.03 packs per capita.
2. **The factor model adds confidence intervals** (via Proposition 1), which the standard synthetic control method lacks.
3. **Structural break confirmed:** Both the Chow test ($T_0 = 1989$) and the QLR sup-F test

(unknown break, 15% trimming) strongly reject $H_0 : \lambda(1) = \lambda(0)$ ($p = 0.000$), validating the causal identification assumption.

4. **Implementation is simple:** 3 steps requiring only PCA + two OLS regressions, with no constrained optimization (unlike the synthetic control).
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11 References

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